

HW 2 P3.4 Demand Function

CPC demand function $Q = 5,000 - 4,000P + 0.02P_{op} + 0.25I + 1.5A$ Given

$P \Rightarrow -4000$. When price increases by \$1 Q -demand decreases by 4000 units, holding everything else constant.

$P_{op} \Rightarrow$ population coefficient is 0.02. Larger "pop" increases demand.

$I \Rightarrow$ income $\Rightarrow 0.25 \Rightarrow$ Higher disposable income increases demand assuming other factors remain constant.

$A \Rightarrow 1.5 \Rightarrow$ Additional advertising demand.

So A $P = 10 \$$ $P_{op} = 1,000,000$ $I = 60,000$ $A = 10,000$

Step 1 substitute values

$$Q = 5,000 - 4,000(10) + 0.02(1,000,000) + 0.25(60,000) + 1.5(10,000)$$

$$Q = 5,000 - 40,000 + 20,000 + 15,000 + 15,000$$

$$Q = 15,000 \text{ coupon books.}$$

Part B

Advertising increase from 10K to 15K

$$A = 15K$$

$$Q = 5,000 - 40,000 + 20,000 + 15,000 + 22,500$$

$$A = 1.5(15,000) = 22,500$$

$$Q = 22,500 \text{ coupon books}$$

Part cChange in demand

$$Q_A = 15,000$$

$$Q_B = 22,500$$

$$\Delta Q = Q_B - Q_A = 7,500 \text{ coupon books}$$

Advertising expenditure increased by \$5,000 causing demand to increase by 7,500 coupon books at the same price. This is the shift in the demand curve. Not movement.

algebraic step: need to express Q as a function of P

$$Q_A = 5,000 - 4,000P + 20,000 + 15,000 + 15,000$$

$$Q_A = 55,000 - 4,000P$$

$$Q_B = 62,500 - 4,000P \quad \underline{P=10}$$

$$P=10$$

$$Q_A = 15,000$$

$$Q_B = 22,500 \text{ matches}$$

Conclusion: The two demand curves are parallel because the price coefficients remains -4000. Increasing advertisement shifts the demand curve to the right by 7,500 coupon books at every given price assuming the linear model remains valid.

HW2 P3.5

$$\text{Supply function} = -59,000,000 + 500,000P - 125,000P_L - 500,000P_K + 2,000,000W$$

Part A $P_L = 8$ $P_K = 10$ $W = 20$

1) $Q = f(P)$
 $P = f(Q)$

$$Q = -59,000,000 + 500,000P - 125,000(8) - 500,000(10) + 2,000,000(20)$$

Part B

$P = \$50, \$60, \$70$

$P_1 = \$50$

$$Q = -25,000,000 + 500,000(50)$$

$Q_1 = 0$ cases

$P_2 = \$60$

$$Q = -25M + 500K(60)$$

$Q_2 = 5,000,000$ cases

$P_3 = \$70$

$Q_3 = 10,000,000$ cases

Conclusion: The higher the quantity the industry must supply, the higher the price required to induce production. Assuming labor costs, capital costs, & rainfall remain unchanged.

$$Q = -25,000,000 + 500,000P$$

$$Q + 25,000,000 = 500,000P$$

$$P = \frac{Q + 25,000,000}{500,000} = 50 + \frac{Q}{500,000}$$

Part C

Supplies = 4M, 6M, 8M

$P = 50 + \frac{Q}{500,000}$ $Q = 4M$
 $Q = 6M$

$$P = 50 + \frac{4,000,000}{500,000}$$

$$P = 50 + 8 = \$58 \quad Q_{4M}$$

$$P = 50 + 12 = \$62 \quad Q_{6M}$$

$$P = 50 + 16 = \$66 \quad Q_{8M}$$

P3.9 Demand curves (Eastern Shuttle, Inc)

Given monthly demand function $Q = 26,000 - 500P - 250P_{OG} + 200I_B$

Variable S captures the seasonal effect.
- 5000 winter ; 5000 summer.

~~Step substitute.~~

January is winter month so $S=0$

Other goods price index $P_{OG} = 4$

Business Activity index $I_B = 250$

$$Q = 26,000 - 500P - 250(4)$$

$$+ 200(250) - 5,000(0)$$

$$Q_{\text{winter}} = 75,000 - 500P$$

Part B summer demand

$$S=1$$

$$Q = 26,000 - 500P - 250(4) + 200(250) - 5,000(1)$$

$$Q_{\text{summer}} = 70,000 - 500P$$

$$P = 100 \$$$

$$Q_{\text{summer}} = 70,000 - 500(100)$$

$$Q = 20,000 \text{ passengers per month.}$$

Total Revenue

$$TR = P \times Q$$

$$= 100 \times 20,000$$

$$= 2,000,000 \text{ per month}$$

summer

$$Q_{\text{winter}} = 75,000 - 500(100)$$

$$Q_{\text{winter}} = 25,000$$

Q winter Revenue

$$TR = P \times Q$$

$$= 100 \times 25,000$$

At the same price 100 \$ ticket, the model

predicts 5,000 fewer passengers in summer than in winter. $= 2,500,000$

Shift in demand curve caused by the seasonal dummy variable rather than movement along the curve caused by price change.

P 3.10 Demand & Supply Curves.

$$Q_D = 317,500 - 10,000P$$

$$Q_S = 2500 + 7500P$$

$$\text{Surplus or shortage} = Q_S - Q_D$$

If $Q_S - Q_D > 0$, there's a surplus

$Q_S - Q_D < 0$, there's a shortage

$Q_S - Q_D = 0$, the market is in equilibrium.

$$Q_S = 2500 + 7,500(15)$$

$$Q_S = 2500 + 112,500 = 115,000$$

$$Q_D = 317,500 - 10,000(15)$$

$$= 317,500 - 150,000 = 167,500$$

$$Q_S - Q_D = 115,000 - 167,500 = -52,500 \text{ (shortage)}$$

Price	Q_S	Q_D	$Q_S - Q_D$	Market condition
15	115,000	167,500	-52,500	shortage
16	122,500	157,500	-35,000	shortage
17	130,000	147,500	-17,500	shortage
18	137,500	137,500	0	Equilibrium
19	145,000	127,500	+17,500	surplus
20	152,500	117,500	+35,000	surplus.